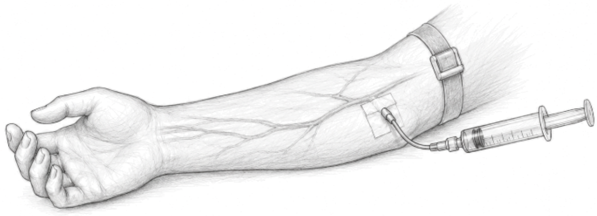
and recombination, RNA maturation, and cellular defense against foreign genetic material. For example, 3′–5′ exonucleases such as DnaQ perform proofreading during DNA replication, while RNase H removes RNA primers from DNA–RNA hybrids to ensure proper genome duplication (Yang, 2011). In DNA repair, nucleases like Mre11 initiate double-strand break processing, and structure-specific nucleases such as FEN1 remove flap intermediates during lagging-strand synthesis (Yang, 2011). In RNA biology, nucleases including RNase E and RNase Z are essential for RNA processing and maturation, whereas Argonaute proteins mediate RNA interference by cleaving target mRNAs (Yang, 2011). Additionally, restriction endonucleases and CRISPR-associated nucleases provide defense against invading genetic elements, highlighting the diverse yet indispensable roles of nucleases across biological systems (Yang, 2011).

CRYSTAL STRUCTURE OF HUMAN FLAP ENDONUCLEASE FEN1 (D181A) IN COMPLEX WITH SUBSTRATE 5'-FLAP DNA AND K+; HUMAN FLAP ENDONUCLEASE STRUCTURES, DNA DOUBLE-BASE FLIPPING, AND A UNIFIED UNDERSTANDING OF THE FEN1 SUPERFAMILY. TSUTAKAWA, S.E., CLASSEN, S., CHAPADOS, B.R., ARVAI, A.S., FINGER, L.D., GUENTHER, G., TOMLINSON, C.G., THOMPSON, P., SARKER, A.H., SHEN, B., COOPER, P.K., GRASBY, J.A., TAINER, J.A. (2011) CELL 145: 198-211

Antisense oligonucleotides (ASOs) inevitably encounter nucleases immediately upon administration and cellular uptake. In the extracellular space and circulation, they are exposed to DNase I, a nonspecific endonuclease that cleaves both single- and double-stranded DNA into short oligonucleotides as part of normal extracellular DNA clearance and turnover. Following cellular uptake via endocytosis, ASOs enter endosomal and lysosomal compartments, where they encounter degradative enzymes including DNase II, and RNases involved in nucleic acid recycling. Within the intracellular environment, additional nucleases such as 3′ exonucleases and endonucleases further metabolize oligonucleotides into shortened fragments, contributing to their overall clearance and activity profile (Jo et al., 2023). For single stranded ASOs that enter the cell nucleus there are numerous nucleases involved in managing various aspects of genomic stability and replication (e.g. FEN1 see figure above). This widespread exposure to therapeutic-damaging enzymes highlights the need for protective strategies when attempting to deliver ASOs to their intracellular targets.



Systemic dosing

Systemic dosing of RNA therapeutics is governed by delivery, chemical modification, tissue targeting and productive intracellular uptake rather than by plasma exposure alone. Naked siRNA, PMO and related oligonucleotide therapeutics are large, highly polar and often negatively charged, which limits passive membrane permeability and makes them vulnerable to nuclease degradation or rapid clearance. As a result, systemically administered RNA therapeutics generally require chemical stabilization, conjugation or delivery systems such as lipid nanoparticles, ligand-directed constructs, cell-penetrating peptides or antibody–oligonucleotide conjugates to protect the RNA, promote tissue uptake and enable intracellular activity (Choi et al., 2025).

Route of administration can reshape the tissue PK/PD profile from the start. In one study (Bäckström et al.), subcutaneous dosing of non-conjugated Malat1 LNA ASO produced the highest concentrations in kidney and liver, whereas intratracheal dosing shifted the highest exposure to lung, followed by kidney and liver. Intratracheal dosing achieved similar lung knockdown at a much lower dose than subcutaneous dosing, illustrating how local or organ-directed administration can reduce the dose requirement for the target tissue while changing systemic exposure patterns. Systemic exposure after non-IV routes depend heavily on route, dose and tissue transit into circulation (Bäckström et al., 2024).

Chemistry is central to the PK/PD behavior of oligonucleotide therapeutics. Phosphodiester linkages are highly susceptible to nuclease degradation, so many antisense oligonucleotides use phosphorothioate backbones to improve nuclease resistance and reduce renal clearance through increased plasma protein binding. However, phosphorothioate modification can reduce target RNA affinity, so ribose modifications such as

2′-O-methoxyethyl, locked nucleic acid and constrained ethyl are used to improve binding affinity and potency. Gapmer ASOs typically combine a phosphorothioate backbone with chemically modified “wings” flanking a central DNA gap that supports RNase H1-mediated target RNA cleavage (Bäckström et al., 2024). PMO chemistries, by contrast, are neutrally charged morpholino oligomers used frequently for splice modulation, including exon skipping in Duchenne muscular dystrophy, but their clinical effect has historically been limited by inefficient delivery to skeletal and cardiac muscle (Yun et al., 2025; Boucher et al., 2025).

Plasma protein binding is another important modifier of systemic oligonucleotide disposition because it can influence circulation time, renal clearance, tissue distribution and the unbound drug fraction available for distribution. Protein binding is strongly chemistry dependent. In a study comparing sequence-matched antisense oligonucleotides, MOE/phosphorothioate-modified ASOs showed much higher plasma protein binding than phosphorodiamidate morpholino oligomers. At 1 μM in human plasma, the unbound fraction was only 4.4% for 20mer-MOE and 1.0% for 25mer-MOE, compared with 56% for 20mer-PMO and 36% for 25mer-PMO (Yun et al., 2025). These data show that backbone and sugar chemistry can materially alter systemic exposure and distribution, even when sequence and length are held constant.

This has direct implications for systemic dosing. Highly protein-bound oligonucleotides may have reduced filtration and slower renal elimination, supporting longer systemic persistence and greater opportunity for tissue uptake. By contrast, weakly protein-bound chemistries such as PMOs may remain more freely circulating and be more readily cleared unless conjugation improves tissue delivery. Yun et al. also found that MOE/PS ASOs showed evidence of saturable plasma protein binding above approximately 1 μM, whereas PMO ASOs did not show saturation up to 10 μM. At clinically relevant micromolar exposures, saturation of binding could increase the unbound fraction, alter tissue uptake and change the apparent PK/PD relationship (Yun et al., 2025).

The identity of plasma binding proteins may also matter. Although albumin is often assumed to dominate oligonucleotide binding, Yun et al. found that human γ-globulins made the largest contribution to binding for both tested PMO and MOE/PS ASOs. For 25mer-PMO, γ-globulins accounted for 58% of binding contribution, compared with 22% for albumin and 20% for α1-acid glycoprotein. For 25mer-MOE, γ-globulins contributed 50%, with albumin contributing 13%, LDL 18%, HDL 17% and α1-acid glycoprotein 2% (Yun et al., 2025). This suggests that disease states associated with altered immunoglobulin levels, such as chronic inflammatory disease, liver disease, cancer or hypergammaglobulinaemia, could plausibly affect systemic oligonucleotide PK and PD.

Plasma concentrations may fall relatively quickly after systemic dosing, but the pharmacological effect can persist because intracellular RNA remains active after uptake into the target tissue. In non-clinical studies, intravenously administered lumasiran in monkeys had a plasma half-life of approximately 0.6 hours, whereas the liver tissue half-life was about 296 hours. Similarly, after IV administration of givosiran in rats, the plasma half-life was approximately 0.2 hours, while the liver half-life was 55 hours and the kidney half-life was 119 hours. Bäckström et al. similarly reported prolonged ASO tissue half-lives in mouse tissues, with non-conjugated Malat1 LNA ASO after subcutaneous administration showing tissue half-lives of approximately 152 hours in liver, 161 hours in kidney, 188 hours in heart and 179 hours in lung. These findings show that plasma exposure captures only the early distribution phase, while tissue retention and intracellular activity better reflect the duration of pharmacological effect (Bäckström et al., 2024; Choi et al., 2025).

These chemistry, conjugation and disposition differences have practical consequences for systemic dosing. Bäckström et al. compared Malat1-targeting ASOs with LNA, cEt and 2′-MOE ribose chemistries after subcutaneous and intratracheal administration in mice. After systemic subcutaneous dosing, tissue distribution was broadly similar across the three chemistries, with the highest concentrations generally observed in kidney, followed by liver, lung and heart. However, the degree of knockdown differed: LNA and cEt produced broadly similar Malat1 knockdown, whereas 2′-MOE generally produced less knockdown in liver, kidney, lung and heart despite measurable tissue exposure. This highlights that tissue concentration alone does not fully predict pharmacodynamic activity, because productive intracellular uptake and intrinsic potency also matter (Bäckström et al., 2024).

The disconnect between tissue concentration and pharmacodynamic effect is a recurring feature of RNA therapeutics. In the mouse study, kidney exposure after subcutaneous dosing was higher than liver exposure, yet liver knockdown was generally greater. This suggests that total tissue concentration includes both productive and non-productive uptake, and that the relevant pharmacological driver is not simply how much oligonucleotide enters a tissue, but how much reaches the correct cell type and intracellular compartment where it can engage the target RNA (Bäckström et al., 2024). The same principle applies to siRNA therapeutics, where the active antisense strand must escape the endosome, enter the cytosol, and load into RISC before durable target mRNA degradation occurs (Jing et al., 2023; Choi et al., 2025).

Tissue targeting can be a dominant determinant of response for systemically dosed RNA therapeutics. In liver-directed siRNA and ASO programs, GalNAc conjugation works not because it is the most powerful endosomal escape technology, but because it efficiently delivers oligonucleotides to hepatocytes through the asialoglycoprotein receptor. Similarly, in DMD, TfR1-targeted antibody–oligonucleotide conjugation may be more impactful than a generalized endosomal escape strategy because it addresses a central delivery barrier: broad, efficient and repeatable delivery to skeletal and cardiac muscle (Bäckström et al., 2024; Boucher et al., 2025).

GalNAc-conjugated siRNAs provide a useful benchmark for this delivery-first principle. GalNAc binds with high affinity and specificity to the asialoglycoprotein receptor, which is highly abundant on hepatocytes and undergoes rapid recycling. After subcutaneous administration, GalNAc–siRNA conjugates are absorbed into systemic circulation and rapidly taken up by hepatocytes through ASGPR-mediated endocytosis. Human plasma exposure is brief, with typical Tmax values of approximately 3–4 hours and plasma half-lives generally under 10 hours, but pharmacodynamic effects can persist for weeks to months because the relevant driver is not plasma concentration or even total liver concentration. Instead, durable activity depends on the small fraction of intracellular siRNA that escapes endosomal trapping, enters the cytosol and loads into RISC.

Preclinical ADME data reinforce this distinction between exposure and productive pharmacology. GalNAc–siRNAs show extensive liver distribution, with liver-to-plasma ratios of approximately 5500 in rats and 6300 in monkeys, and recovery of up to 63% of dose in rat liver and 87% in monkey liver after pharmacologically relevant subcutaneous doses. Yet even in the liver, total tissue concentration does not fully explain the pharmacodynamic response. RISC-loaded siRNA better tracks the timing and durability of target mRNA knockdown, helping explain why GalNAc–siRNAs can show transient plasma PK but prolonged pharmacology. This disconnect between short systemic exposure and durable intracellular activity supports infrequent dosing schedules, including monthly dosing for givosiran and lumasiran, every three-month dosing for vutrisiran, and twice-yearly maintenance dosing for inclisiran.

The same distinction is evident with lipid nanoparticle-mediated siRNA delivery. The clearest clinical example of systemic intravenous siRNA delivery remains patisiran, an RNA interference therapeutic formulated in lipid nanoparticles and administered by IV infusion every three weeks. Unlike GalNAc-conjugated siRNAs, which are generally given subcutaneously and rely on receptor-mediated uptake, patisiran uses an LNP formulation to support systemic administration and liver delivery. Following IV administration, LNPs accumulate predominantly in the liver, partly through interactions with serum apolipoprotein E and uptake by hepatocytes via low-density lipoprotein receptors (Choi et al., 2025).

Patisiran demonstrates how systemic dosing can produce durable pharmacodynamic effects despite complex disposition. After 0.3 mg/kg IV administration every three weeks in patients, patisiran had a reported plasma half-life of approximately 76.8 hours, with a Cmax of 7150 ng/mL and AUC of 184,000 ng·h/mL. At steady state, the plasma half-lives of the LNP components DLin-MC3-DMA and PEG2000-C-DMG were approximately 10.9 and 3.76 days, respectively. Clinically, this exposure supported around an 80% reduction in serum transthyretin, demonstrating that systemic LNP delivery can translate into substantial and sustained target protein knockdown (Choi et al., 2025). Here again, plasma PK is not the main determinant of pharmacologic durability: the relevant sequence is liver deposition, hepatocyte uptake, endosomal escape, RISC loading and sustained target engagement.

Together, GalNAc conjugates and LNPs show that successful RNA pharmacology depends on more than systemic exposure. GalNAc has been successful because it solves the upstream targeting problem for hepatocytes: reliable, receptor-mediated delivery to the right liver cell type at sufficient scale. LNPs solve a related problem through formulation-driven liver tropism after IV administration. In both cases, productive pharmacology depends on the fraction of delivered oligonucleotide that reaches the appropriate intracellular compartment, not simply on the amount present in plasma or even in bulk tissue.

The recent clinical contrast between Entrada Therapeutics and Avidity Biosciences in Duchenne muscular dystrophy is consistent with this broader delivery principle. Entrada's ENTR-601-44 is an Endosomal Escape Vehicle-conjugated PMO designed for exon 44 skipping. In the first cohort of the Phase 1/2 ELEVATE-44-201 study, participants received three intravenous 6 mg/kg doses, and the company reported favorable safety and tolerability, but lower-than-expected plasma Cmax and AUC in pediatric DMD participants compared with healthy adult volunteers. The molecular readout was also modest relative to expectations: ENTR-601-44 increased dystrophin by 2.36% over baseline, below the double-digit dystrophin production that investors and analysts had anticipated (Entrada Therapeutics, 2026; Taylor, 2026).

By contrast, Avidity's delpacibart zotadirsen, or del-zota, is an antibody–oligonucleotide conjugate that links an exon 44-skipping PMO to an antibody targeting transferrin receptor 1, a receptor expressed on skeletal and cardiac muscle. In the Phase 1/2 EXPLORE44 study, Avidity reported skeletal muscle PMO concentrations of 200 nM and approximately 25% normal dystrophin production after three doses. The company also reported greater than 80% reductions in creatine kinase compared with baseline, consistent with a biologically meaningful reduction in muscle damage (Avidity Biosciences, 2025).

Together, these data suggest that endosomal escape is necessary but not sufficient. Once an oligonucleotide is internalized, escape into the cytosol or nucleus remains critical. However, the Entrada and Avidity data imply that, in DMD, an upstream delivery bottleneck may be more important: getting enough drug into the right skeletal and cardiac muscle cells in the first place. A platform optimized for cellular uptake and endosomal escape may still underperform if systemic exposure is lower than expected, if delivery to muscle is inefficient, or if a large fraction of total uptake is non-productive. By contrast, Avidity's TfR1-directed antibody conjugation may more directly address the upstream delivery problem by enriching PMO in skeletal and cardiac muscle, increasing the probability that enough oligonucleotide reaches the relevant intracellular compartment to drive exon skipping and dystrophin restoration (Avidity Biosciences, 2025; Boucher et al., 2025).

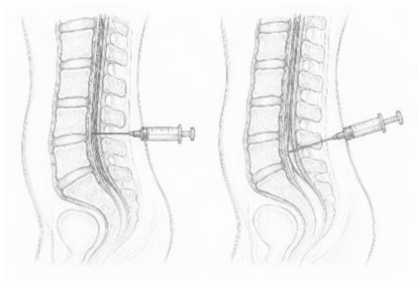
This delivery distinction becomes even more important as RNA therapeutics move toward larger and more structurally complex cargos. Short siRNAs, splice-switching ASOs, PMOs, mRNAs and gene-editing systems all face the same broad delivery problem, but they impose different intracellular requirements. A small siRNA must reach the cytosol and load into RISC; a splice-modulating oligonucleotide often needs access to nuclear pre-mRNA; an mRNA must escape the endosome intact and remain translationally competent; and gene-editing cargos may require coordinated delivery of larger nucleic acid, protein or ribonucleoprotein components. As cargo size and complexity increase, total tissue exposure becomes an even less reliable proxy for productive pharmacology. The key question shifts from whether a platform can deliver oligonucleotide mass to a tissue, to whether it can deliver the right molecular cargo, in the right form, to the right intracellular compartment, at sufficient levels to drive the intended biology.

For PK/PD modelling, systemic RNA therapeutics therefore require approaches that go beyond conventional plasma exposure-response relationships. Models need to account for plasma distribution, protein binding, unbound fraction, target-organ uptake, conjugation chemistry, receptor-mediated targeting, endosomal escape, intracellular RISC or RNase H1 engagement, target RNA degradation and downstream biomarker change. This is why liver or RISC compartments have been incorporated into PK/PD models for several siRNA therapeutics, including givosiran and lumasiran, while patisiran has been modelled using plasma and liver compartments linked to an indirect pharmacodynamic model of serum transthyretin reduction (Choi et al., 2025). For ASOs and PMOs, clinical experience in DMD now suggests that models should explicitly capture muscle delivery and productive uptake, not just plasma exposure or theoretical endosomal escape efficiency (Entrada Therapeutics, 2026; Avidity Biosciences, 2025).

Systemic dosing also has important safety implications. IV lipid-based formulations can trigger infusion-related reactions or innate immune activation, requiring attention to formulation composition, premedication and tolerability. Patisiran, for example, requires premedication to reduce infusion-related reactions. More broadly, systemic delivery can expose non-target tissues such as kidney, spleen and immune cells, so biodistribution, off-target hybridisation, immune stimulation, organ-specific accumulation, plasma protein binding and chemistry-specific tissue retention should be evaluated alongside efficacy biomarkers (Jing et al., 2023; Choi et al., 2025; Yun et al., 2025).

Overall, the PK/PD of systemically dosed RNA therapeutics is best understood as a multi-step delivery and disposition process that begins with administration and systemic exposure, passes through plasma binding, tissue distribution, tissue targeting, cellular uptake and intracellular trafficking, and ends with target engagement and durable pharmacodynamic response. Plasma PK describes systemic exposure, but pharmacodynamic durability depends on how much RNA remains protected in circulation, how it interacts with plasma proteins, how much reaches the target organ, how chemistry and conjugation affect productive uptake, and how long the active intracellular species persists. The Entrada and Avidity DMD data sharpen this conclusion: enhancing endosomal escape may help, but the more decisive variable may be tissue-targeted delivery to the right disease-relevant cells. In systemic RNA therapeutics, the winning platform is likely to be the one that delivers sufficient active oligonucleotide to the correct tissue first, and only then solves intracellular release and target engagement (Jing et al., 2023; Bäckström et al., 2024; Choi et al., 2025; Yun et al., 2025; Avidity Biosciences, 2025; Entrada Therapeutics, 2026).

Intrathecal administration



In 1846, William T.G. Morton demonstrated anesthesia, allowing surgeons to perform operations without causing severe pain; a revolutionary medical advancement that marked the beginning of modern anesthesiology. In 1898, August Bier administered cocaine into the intrathecal space of a patient, creating the first successful spinal anesthetic. His pioneering work laid the foundation for modern regional anesthesia and inspired further developments in both spinal and epidural techniques. Shortly thereafter, in 1901, the Romanian surgeon Racoviceanu-Pitesti reported the first use of opioids administered intrathecally, recognizing their potential to enhance analgesia through neuraxial pathways. In 1979, Behar and colleagues published the first report describing epidural morphine for pain management, demonstrating prolonged and effective analgesia. These discoveries transformed perioperative pain control and led to the widespread adoption of neuraxial opioids as part of routine intraoperative and postoperative analgesic regimens.

The historical progression of neuraxial anesthesia also established the intrathecal space as a valuable route for therapeutic delivery beyond anesthetics and analgesics. As understanding of cerebrospinal fluid dynamics, spinal anatomy, and drug distribution improved, clinicians and researchers increasingly recognized the potential of intrathecal administration to bypass systemic barriers and achieve targeted effects within the central nervous system. This concept has become particularly important in the field of genetic medicine. The blood-brain barrier, while essential for protecting neural tissue, presents a significant obstacle to the delivery of many biologic therapies, including viral vectors, nucleic acids, and gene-editing technologies. Intrathecal administration offers a means of circumventing this barrier by delivering therapeutic agents directly into the cerebrospinal fluid, thereby facilitating broader access to the brain and spinal cord.

Today, intrathecal delivery has emerged as a promising platform for gene replacement, gene silencing, and gene-editing therapies aimed at treating previously intractable neurological disorders. The foundations of this approach can be traced to the early development of adeno-associated virus (AAV) vectors for central nervous system (CNS) gene transfer. In 1994, Kaplitt and colleagues demonstrated long-term transgene expression and phenotypic correction in the mammalian brain following AAV-mediated gene transfer, providing some of the earliest evidence that genetic therapies could be delivered directly to neural tissues (Kaplitt et al., 1994). Subsequent studies confirmed persistent CNS transduction and tissue-specific expression patterns following AAV administration, establishing the feasibility of long-term gene expression within the nervous system (McCown et al., 1996). These advances ultimately led to the first clinical applications of CNS-directed gene therapy, including treatment of Canavan disease, a fatal leukodystrophy caused by mutations in the ASPA gene. In a landmark clinical study, AAV2-mediated gene transfer resulted in reduced levels of N-acetylaspartate, slower progression of cerebral atrophy, and stabilization of neurological function, demonstrating the therapeutic potential of direct CNS gene delivery (Leone et al., 2012).

More recently, the success of intrathecally administered antisense oligonucleotides for spinal muscular atrophy (SMA) has validated the cerebrospinal fluid as a route for repeated genetic intervention. Nusinersen, an antisense oligonucleotide that modifies SMN2 splicing to increase production of survival motor neuron protein, significantly improved motor function and survival in patients with SMA and established a new paradigm for neuraxial genetic medicine (Finkel et al., 2017). Building upon this success, intrathecal and intracisternal delivery approaches are being explored for a growing number of neurological disorders. In amyotrophic lateral sclerosis, tofersen has demonstrated the feasibility of suppressing expression of mutant SOD1 through repeated intrathecal administration (Miller et al., 2022). Similarly, AAV-mediated gene replacement strategies have been investigated for giant axonal neuropathy, where intracisternal administration of AAV9 vectors achieved widespread CNS distribution and substantial pathological improvement in preclinical models (Mussche et al., 2013).

Endosomal Escape

Early studies using electron microscopy with gold-labelled siRNA enabled direct visualization and counting of intracellular particles, revealing that the vast majority localize within endosomes, with only rare cytosolic events, corresponding to ~1–2% escape during a narrow window of